Faizan Farhad

AI / ML Engineering Intern

faizanfarhad2211@gmail.com | +91 9310905797 | Narela, Delhi

LinkedIn | GitHub | Portfolio

PROFILE

BCA student focused on AI/ML and AI engineering, with hands-on experience building LLM-powered agents, NLP applications, and AI-driven data tools. Experienced in developing natural-language-to-SQL systems, browser automation workflows, and data analysis applications. Interested in building reliable AI systems that solve practical problems through automation, intelligent reasoning, and tool integration.

EDUCATION

Bachelor of Computer Applications (BCA)
CPJ College 2024 – 2028(Expected) | Narela, Delhi

SKILLS

- **Languages**: Python, SQL
- **AI / ML & Agents:** PyTorch, Hugging Face Transformers, BitsAndBytes, Scikit-Learn, Ollama, Prompt Engineering, Agentic Workflows
- **Backend & Web:** FastAPI, Uvicorn, REST APIs, HTTPX, SQLAlchemy, OAuth 2.0
- **NLP:** Text Preprocessing, Tokenization
- **Deep Learning**: PyTorch, CNN, Neural Networks
- **Databases:** SQLite

LANGUAGES

- Hindi
- English

PROJECTS

Browser Automation Agent [Sep 2026 – Oct 2026]

• Engineered an LLM-driven browser automation agent that converts natural-language tasks into structured action plans and executes them through Playwright.
• Designed a planner–validator–executor architecture with bounded replanning, validating generated actions against live browser state before execution and using DOM observations to recover from invalid plans.
• Engineered an autonomous LLM agent executing structured web tasks through an API-first router, executing direct HTTP requests for read operations and Playwright for complex DOM interactions.
• Integrated Chrome DevTools Protocol (CDP) for persistent session attachment and desktop Google OAuth 2.0 to automate authenticated user workflows (e.g., Gmail).
• Implemented runtime state reliability via atomic JSON snapshots (os.replace), logging step-by-step attempt histories, failed DOM observations, and execution traces
Tech Stack: Python, Playwright, CDP (Chrome DevTools Protocol), Ollama, DeepSeek API, HTTPX, OAuth 2.0

Autonomous CLI Agent & Episodic Memory System [Oct 2026 – Nov 2026]

• Architected an extensible CLI terminal agent featuring dedicated tool registries for safe OS operations, sandboxed bash commands, and direct file-system manipulation within Dockerized environments.
• Engineered a custom RAG-based episodic memory engine (relevantMemoryFetch) integrating document cleaning, sliding-window chunking, vector embeddings, and top-k semantic retrieval to persist user context across multi-turn sessions.
• Implemented a two-stage retrieval and re-ranking pipeline over long-term markdown logs (savedMem), filtering irrelevant execution context and minimizing token bloat during complex reasoning tasks.
• Designed an autonomous execution loop with structured parameter parsing, command error interception, and automated tool dispatching to streamline complex terminal workflows from natural-language prompts.
Tech Stack: Python, Docker, RAG, Semantic Embeddings, OS/File Tools, Prompt Engineering, Vector Retrieval, LLM Tool Calling

COURSES

AI/ML | Daksh Gurukul (IIT Guwahati)
June 2025 – Oct 2026

• Acquired deep theoretical foundations and practical applications in AI/ML principles, deep learning frameworks, and predictive modeling techniques to solve real-world problems.